

Minimum Viable Technical Specification (MVTs)

on development of VialApp platform

1. Problem and target user

Problem:

1. People in Chile who are preparing for the theoretical driving test need an easy way to practice before taking the official exam. VialApp is designed to provide practice tests and study tools online, so users can prepare from their computer or mobile device.

Target user

2. The target audience is people in Chile who are preparing to obtain a Class B or Class C driving license and need to study for the theoretical driving test. VialApp is mainly intended for people who want to practice online and evaluate their knowledge before taking the official test.

2. Core User Scenarios

Scenario 1 – Try a Free Practice Test

A visitor enters VialApp and starts a free practice test without purchasing a subscription. The user answers the questions and receives the result after completing the test.

Scenario 2 – Create an Account and Log In:

A user creates an account using the registration form and then logs in to VialApp. After logging in, the user can access their account and see their subscription status.

Scenario 3 – Purchase a Subscription:

A registered user selects a 7, 15, or 30-day subscription plan and continues to the payment process. After the payment is confirmed, the system automatically activates the subscription and gives the user access to the course.

Scenario 4 – Complete a Practice Test and Review Results:

A user completes a practice test and receives a final score. The user can review which answers were correct and incorrect to identify the topics that need more practice.

3. Functional Requirements

FR-01 – Free Practice Test

The system shall allow visitors to take a limited number of practice tests without purchasing a subscription.

FR-02 – User Registration

The system shall allow users to create an account using a registration form.

FR-03 – User Login

The system shall allow registered users to log in and access their account information and subscription status.

FR-04 – Course Selection

The system shall allow users to select between the Class B and Class C driving license courses.

FR-05 – Subscription Plans

The system shall display the available 7, 15, and 30-day subscription plans, including their duration and price.

FR-06 – Online Payment

The system shall allow a registered user to pay for a selected subscription plan using the online payment system.

FR-07 – Automatic Subscription Activation

The system shall activate the user's subscription automatically after the payment has been successfully confirmed by the payment system.

FR-08 – Practice Test Completion

The system shall allow users to answer practice test questions, move between questions, and submit the test when finished.

FR-09 – Test Results

The system shall show the user's final score and allow them to review their correct and incorrect answers after completing a test.

FR-10 – Password Recovery

The system shall allow registered users to recover access to their account if they forget their password.

FR-11 – Subscription Expiration

The system shall automatically expire a subscription after its 7, 15, or 30-day duration. A user can purchase a new subscription after the previous subscription expires.

4. Non-Functional Requirements

NFR-01 – Responsive Design

The platform should work correctly on desktop, tablet, and mobile devices.

NFR-02 – Browser Compatibility

The platform should support the main modern browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

NFR-03 – Security

The platform should use HTTPS for communication between the frontend, backend, and external services. User authentication and payment information should be handled securely.

NFR-04 – Usability

The interface should be simple and easy to understand, especially for users taking practice tests or purchasing a subscription.

NFR-05 – Performance

The main pages and practice tests should load without unnecessary delays under normal usage conditions.

5. Acceptance Criteria

AC-01 – User Registration

Given that a new user is on the registration page, When the user enters valid information and submits the form, Then the system should create the account and allow the user to log in.

AC-02 – User Login

Given that a registered user has an account, When the user enters the correct login information, Then the system should allow access to the user account.

If the login information is incorrect, the system should show an error message and deny access.

AC-03 – Subscription Purchase and Activation

Given that a registered user has selected a subscription plan, When the payment is successfully confirmed by the payment system, Then the system should automatically activate the selected subscription and give the user access to the course.

If the payment is rejected or cannot be confirmed, the subscription should not be activated.

AC-04 – Practice Test

Given that the user has access to a practice test, When the user answers the questions and submits the test, Then the system should process the answers and generate a final result.

Users should be able to answer questions, navigate through the test, submit their answers, and receive a final result.

AC-05 – Test Results

Given that a user has completed a practice test, When the result is generated, Then the system should show the final score and identify the correct and incorrect answers.

If no completed test exists, the system should not show a result for that test.

6. MVP Scope

In MVP Scope

The first version of VialApp will include the main functions required for users to study and purchase access to the platform:

- User registration and login.
- Free practice tests.
- Class B and C courses.
- 7, 15, and 30-day subscription plans.
- Online payments through Webpay.
- Automatic subscription activation after successful payment.
- Practice test completion and score calculation.
- Review of correct and incorrect answers.
- Basic account information and subscription status.
- Responsive access from desktops and mobile browsers.

Out of MVP Scope

The following features will not be included in the first version:

- Native Android or iOS application.
- Social login with Google, Facebook, or other platforms.
- Advanced statistics or personalized learning recommendations.
- Gamification features such as rankings, achievements, or points.
- Live classes or communication with instructors.
- Multiple payment gateways.
- Advanced administration dashboards.
- Support for driving licenses other than Class B and Class C.
- Automatic subscription renewal, refunds, and cancellation workflows.

7. Constraints, Assumptions and Dependencies

Constraints

- The first version of VialApp will be developed as a web application, not as a native mobile application.
- The backend will be developed with FastAPI and the frontend with React, following the technology stack already defined for the project.
- The project should be completed within the academic period.
- The MVP will focus only on Class B and Class C driving licenses.
- The first version should remain simple and should not include features that are outside the defined MVP scope.
- For the MVP, practice questions and course content will be added directly to the database or through import scripts. A content management interface is not required.

Assumptions

- Users will have access to the Internet and a modern web browser.
- Users who purchase a subscription will have a valid payment method supported by Webpay.

- The study content and practice questions for Class B and Class C will be available and reviewed before publication.
- The owners of VialApp will manage the technical operation, content, and basic administration of the platform.
- Most users will access the platform from desktop or mobile browsers, without needing to install an application.

Dependencies

- VialApp depends on a hosting service to keep the backend and frontend available online.
- The platform depends on Webpay by Transbank to process subscription payments.
- The system depends on a database to store user accounts, subscriptions, test information, and results.
- An email service is required for account notifications and password recovery.
- DNS and domain services are required so users can access the platform through its public domain. These external integrations were already identified in the previous technical specification.

8. Data and External Services Plan

Data

VialApp will store only the information required for the main functions of the platform.

The main types of data are:

- User account information.
- Subscription status and subscription duration.
- Practice test questions and answer options.
- User answers and test results.
- Payment confirmation information.
- Basic account recovery information.

External Services

VialApp will use several external services to support the platform:

- Webpay by Transbank: used to process online payments for subscription plans.
- Email service: used for account notifications and password recovery.
- Hosting service: used to run the frontend, backend, and database.
- DNS and domain service: used to make the platform available through its public domain.

These integrations were already identified in the previous technical specification. For security, API keys, credentials, and configuration variables should not be stored directly in the public source code. They should be managed separately as environment variables or private configuration values. This also matches the previous requirement that external services will need credentials, tokens, and configuration variables.

9. Risks and Open Questions

Main Risks

Risk 1 – Payment Integration Problems

If the integration with Webpay does not work correctly, users may not be able to complete a purchase, or the subscription may not be activated after payment. Since the payment result must be verified by the backend before giving access, this is one of the most important technical risks.

Risk 2 – Incorrect or Outdated Course Content

The practice questions and study content must be reviewed and updated when necessary. If the content is incorrect or outdated, users could prepare information that is no longer valid.

Risk 3 – Problems with External Services

VialApp depends on external services such as hosting, email, DNS, and the payment gateway. If one of these services is unavailable, some functions of the platform may stop working.

Risk 4 – Security Problems

User accounts, authentication data, and payment-related information must be handled securely. Poor management of credentials or configuration variables could create security problems.

Risk 5 – MVP Scope Becomes Too Large

Adding too many features during development could make it difficult to complete the project during the academic period. For this reason, the first version should remain focused on the main functions already defined in the MVP.

Open questions

- How many free practice tests should a visitor be able to complete before registration or subscription is required?
- What email service will be used for notifications and password recovery?
- What hosting provider will be used for the first deployed version?
- How often should the Class B and Class C course content be reviewed and updated?

10. AI Requirements Review

An AI assistant was asked to review the MVTs and identify ambiguities or hidden scope that could make the MVP larger than expected. Three suggestions were received.

AI Suggestion	Decision	Reason
Define exactly how many free practice tests a visitor can complete.	Accept	The original requirement was not specific enough. For the MVP, a visitor will be able to complete one free practice test without registration.
Clarify how questions and course content will be added and maintained, since an administration interface could increase the scope.	Accept	An administration dashboard is not necessary for the MVP. Questions and course content can be added directly to the database or using import scripts.
Define what happens when a subscription expires or when a user wants to purchase another subscription.	Modify	Subscription expiration should be defined, but advanced functions such as automatic renewal, refunds, and cancellation are not required for the MVP. A subscription will expire automatically after 7, 15, or 30 days, and the user can purchase another subscription after it expires.

11. Changes Since Previous Milestone

The main concept of VialApp has not changed since the previous milestone. The platform is still focused on helping users in Chile prepare for Class B and Class C theoretical driving tests, using the same main technologies and payment integration defined before.

The main change in this milestone was to make the project more specific and easier to verify. The previous technical specification described the general structure and functions of the platform, while the new MVTs defines:

- the main user scenarios;
- functional and non-functional requirements;
- acceptance criteria;
- what is included and excluded from the MVP;
- constraints, assumptions, and dependencies;
- data and external services;
- main risks and open questions.

12. Verification

The MVTs were reviewed against the previous technical specification to make sure that the main functions of VialApp are still consistent with the project. The core user scenarios were also checked against the functional requirements and acceptance criteria. The MVP scope was reviewed to avoid adding unnecessary features during development.